Fact-Oriented Data Retrieval via MCTS-Guided Multi-Agent Reasoning with Persistent Knowledge Graphs and Systematic Gap Analysis

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***Abstract*—** In complex information retrieval tasks, it is often necessary to go beyond the task of relevance matching: one has to identify missing information, resolve inconsistencies, and combine information from different sources into a coherent answer. This framework proposes a gap-aware, computationally efficient method of information retrieval by combining three main components:

Strategic Search: Employing Monte Carlo Tree Search (MCTS) to search through information gaps based on priority.

Multi-Agent Coordination: A sequence of tailored agents for planning, retrieval, validation, and synthesis.

Structured Evidence: A knowledge graph that maintains evidence, as well as metrics for its origin and confidence.

The approach recognizes five categories of information gaps: entity, relation, temporal, conflict, and coverage, and plans a series of retrieval actions with priorities to close the gaps with minimal cost. The agents work together: a planner directs the search, retrievers obtain documents, validators determine credibility, and a synthesizer updates the knowledge graph. Conflicts are resolved by weighted aggregation of credibility, and the output includes supported claims, open gaps, and traces of reasoning.

Our approach outperforms baseline approaches to retrieval in terms of consistency, interpretability, and cost-sensitivity.

1 Introduction

Information retrieval systems are increasingly expected to resolve complex, multi-faceted queries that contain implicit assumptions, missing context, or contradictory data. Conventional RAG systems tend to fail when faced with such conditions. The majority of the existing approaches [1, 24] are based on linear single-step retrieval, a technique that assumes the retrieved information is self-consistent and complete. This assumption has significantly constrained their ability to deal with ambiguity and partial knowledge.

Although recent advances in Monte Carlo Tree Search-enhanced retrieval (MCTS-RAG) [2] and multi-agent systems such as MA-RAG [7] and MAIN-RAG [8] have improved factuality through modular reasoning, there are two important bottlenecks that remain:

1. **Lack of Gap-Awareness:** The existing systems do not take into account the information gaps, such as unanswerable entities, time gaps, or fact inconsistencies.
2. **Computational Inefficiency:** They do not focus on the "cost of reasoning," which often results in redundant document retrievals or computations that provide diminishing returns.

To fill these gaps, we introduce a framework that formalizes information retrieval as a cost-aware planning problem with epistemic uncertainty. Our approach combines three major pillars:

- **Gap-Prioritized MCTS Planning:** We rely on a variant of Upper Confidence Bound for Trees (UCT) formulation [4, 5] to concentrate on actions specific to filling high-impact information gaps.
- **Coordinated Multi-Agent Pipeline:** A dedicated set of retrievers, validators, and synthesizers work together through a blackboard system, supporting modular and transparent information processing [3, 7, 8].
- **Persistent Knowledge Graph:** A dynamic knowledge graph [26, 27] that represents facts with temporal information, attribution to sources, and confidence measures, enabling evidence chaining.

The framework identifies gaps in five dimensions: entity, relation, temporal, conflict, and coverage. The framework chooses the retrieval actions based on their expected utility in closing the gaps versus the cost of computation.

Moreover, it computes credibility using a multi-factor scoring function that considers source reputation and confirmation [10, 11].

Ultimately, our system offers more than a final answer; it offers traceable evidence chains and "gap reports" that point out the remaining uncertainties. This framework generalizes existing methodologies by offering a robust, transparent, and cost-effective solution for high-stakes information retrieval.

2 Related Work

The process of factual data retrieval involves understanding the progress made in a number of interrelated fields. We position MCTSRAG-KG in the context of six key areas: retrieval-augmented generation, search planning, multi-agent reasoning, source credibility, information extraction, and reasoning augmentation.

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation [1] is a technique that improves generative models by allowing access to external documents. The standard RAG framework supports single-pass retrieval and answer generation without considering uncertainties or gaps. While dense passage retrieval [24] and reranking strategies [23] have improved the accuracy of document retrieval, they are still stateless and short-sighted in reasoning. More recent approaches such as MCTS-RAG [2] and MA-RAG [7] have mitigated this issue using iterative planning and multi-agent decomposition. Nevertheless, the problem of gap modeling and cost optimization is still not addressed in current approaches.

2.2 Monte Carlo Tree Search (MCTS)

Monte Carlo Tree Search (MCTS) is a method that explores the trade-off between exploration and exploitation using simulated rollouts with the UCT algorithm [4, 5]. MCTS has been applied to retrieval planning [2, 29], allowing the system to choose actions that maximize long-term reward. Our approach builds on this by incorporating reward shaping and cost-aware tree traversal** based on structured search approaches [22].

2.3 Multi-Agent Retrieval Systems

Multi-agent retrieval systems integrate multiple specialized modules to enhance retrieval relevance and robustness. MA-RAG [7] and MAIN-RAG [8] break down tasks among multiple agents that plan, retrieve, and validate together. MAgICoRe [9] proposes iterative coordination among agent roles with coarse-to-fine refinement. Our system builds upon these concepts but incorporates an additional gap management component, enabling agents to coordinate not only around tasks, but also around the uncertainty structure of the query itself.

2.4 Source Credibility and Fact Verification

Factual reliability depends on resolving conflicting evidence. Benchmarks like FEVER [10] and fact classification work [11] highlight the need for source-aware and claim-level verification. Language models can model confidence [19], but few systems integrate credibility scoring directly into retrieval planning. We extend this direction by combining confidence scoring, corroboration, and conflict resolution using structured KG updates and provenance tracking.

2.5 Information Extraction and Entity Linking

Entity and relation extraction are the cornerstones of structured knowledge retrieval. High-quality disambiguation [13] and distant supervision [14] facilitate scalable information extraction. Structured models [15] and KG embeddings [26, 27] are useful in connecting noisy and incomplete information. Our agents apply these techniques to fill a knowledge graph that persists for reasoning and memory, an essential infrastructure for multi-step knowledge retrieval.

2.6 Reasoning Enhancement in Language Models

Recent approaches such as Chain-of-Thought prompting [16], Tree-of-Thoughts [17], and Self-Consistency [18] have enhanced reasoning interpretability and efficiency. Our model applies similar concepts to the retrieval process: gap identification, self-criticism, and answer refinement. Moreover, we incorporate planning-based approaches from Self-Refine [20] and iRe²f [21] to enhance answer selection in multiple iterations. In contrast to these generative approaches, we integrate reasoning into the retrieval process through MCTS-guided exploration.

3 Problem Formulation and Preliminaries

We formalize factual, multi-perspective information retrieval as the problem of producing a consistent and verifiable knowledge fact set $F = \{f_1, f_2, ..., f_{\square}\}$, from diverse sources $S = \{s_1, ..., s_{\square}\}$, for an open-ended natural language query q from a user U. This has to be done in the presence of conflicting information, incomplete evidence, ambiguous entities, and computational resource constraints.

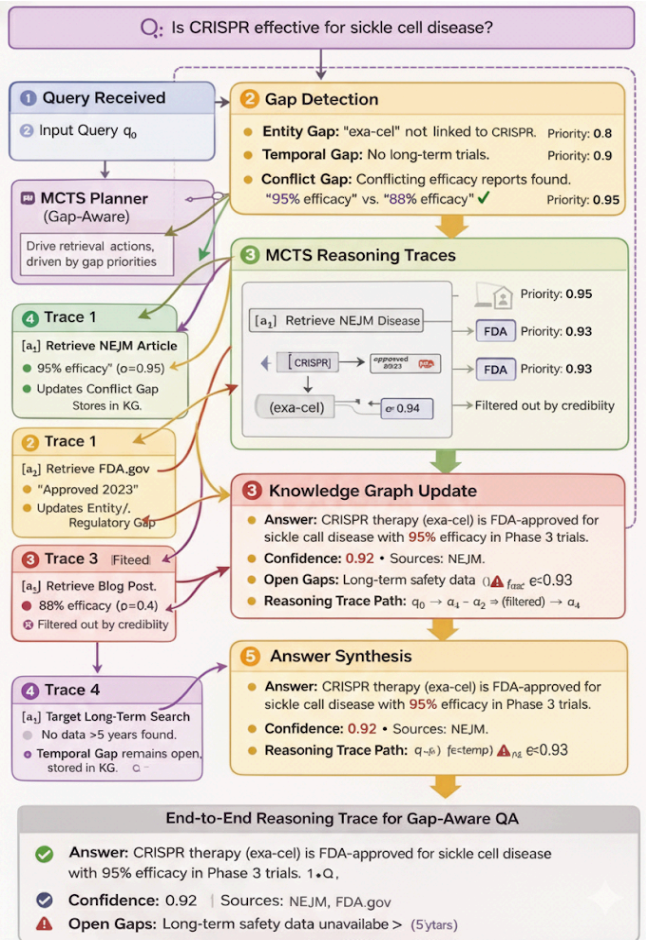

Figure 1: Overview of the MCTSRAG-KG framework. A user query is decomposed by the planner into prioritized information gaps, which guide a gap-aware Monte Carlo Tree Search retrieval process. Specialized agents retrieve, validate, and synthesize evidence while updating a persistent knowledge graph. The final output includes an answer, supporting evidence chains, and an explicit report of unresolved information gaps.

3.1 Objective and Constraints

Input:
A user query q (which may be vague, underspecified, or have latent assumptions).

Goal:
Produce a structured factual answer set F, with reasoning traces, evidence chains, and information gaps.

Constraints:

- **Factual Precision:** Resist hallucinations and illogically inferred facts [1, 2, 3].
- **Gap Awareness:** Identify and prioritize gaps in ambiguous knowledge [2, 7].
- **Conflict Resolution:** Resolve contradictions across sources using credibility and corroboration [10, 11, 12].
- **Computational Efficiency:** Optimize retrieval scope with minimal document access [22, 29].
- **Transparent Reasoning:** Provide traceable evidence chains, decisions, and information gaps [16, 20].

Output:

- A synthesized answer A grounded answer in verifiable facts.

- Evidence chains supporting each fact: (fact → source → credibility)..

- A **gap report**: open questions, priorities, and their retrieval status.
- A **reasoning trace**: sequence of decisions across iterations and agents.

3.2 Gap Taxonomy

We extend the formalism proposed in MCTS-RAG [2] by defining five epistemic types of gaps that affect planning and prioritization:

- **Entity Gaps**: Missing entities required to resolve the query.
  Example: "What therapy treats X?" where "X" is not well defined and ambiguous.

  → Addressed via entity linking and disambiguation techniques. [13], [15].

- **Relation Gaps:** Relations between known entities which are incomplete or absent..
  Example: Knowing that CRISPR and sickle cell exist, but not how it relates.
  → Filled through relation extraction and graph-based expansion [14, 26].

- **Temporal Gaps:** These represent a lack of any knowledge of time-specific anchoring or sequences.
  Example: Comparing approvals in 2023 with results from 2020.
  → Addressed by temporal reasoning modules [27, 29].
- **Conflict Gaps**: Discrepancies within the sources for the exact same fact.
  Example "95% efficacy" vs. "88% efficacy"
  → Resolved through credibility scoring [10, 11,

12].

- **Coverage Gaps:** Answerable Questions with Incomplete Perspectives
  Example: Short-term efficacy is known, long-term risks are missing.
  → Expressed explicitly in the gap report.

These are areas of incompleteness that affect the search planning and reward function of Section 3.4.

3.3 Knowledge Graph Preliminaries

We define a dynamic and persistent **Knowledge Graph (KG)** to track known facts, uncertainty, and evidence provenance.

Formal Definition:

G = (V, E, ℓ)
Where:

- **V = V_entity ∪ V_fact ∪ V_gap** (entity nodes, fact nodes, gap nodes)
- **E ⊆ E_relation ∪ E_evidence ∪ E_gap** (semantic relations, source edges, gap indicators)
- **ℓ: V ∪ E → Labels** with confidence scores, timestamps, or provenance info.

Node Types:

- **Entity Nodes**: `{name, type, attributes, time}`
- **Fact Nodes**: `{subject, predicate, object}` with **confidence ∈ [0,1]**
- **Gap Nodes**: `{gap_type, description, priority_score ∈ [0,1]}`

Edge Types:

- **Relation Edges**: semantic triples with confidence
- **Evidence Edges**: link to source documents **{s_1, ..., s□}** with timestamps and citation scores
- **Gap Edges**: express incomplete paths, used to plan retrieval.

The KG is incrementally updated by synthesis agents and validated by credibility evaluators [7, 9, 10]. It serves both as memory and a reasoning substrate.

3.4 MCTS-Based Planning

We adapt **Monte Carlo Tree Search (MCTS)** [4, 5, 6] to serve as the central planning algorithm for retrieval sequencing.

Each tree node represents the state:
s = {q, D, G}
where:

- **q**: Current query variant
- **D**: Documents retrieved so far
- **G**: Current KG including open gaps

UCT with Gap-Aware Reward:
We extend UCT scoring with a gap-reduction reward:

$$\mathrm{UCT}_{\mathrm{gap}}(n_{ij}) = \frac{Q(n_{ij})}{N(n_{ij})} + c\sqrt{\frac{\ln N(\mathrm{parent})}{N(n_{ij})}} + \alpha \cdot \mathrm{GapScore}(n_{ij})$$

Where:

- **Q(n□)**: cumulative reward based on gap reduction
- **N(n□)**: visit count
- **c**: exploration coefficient
- **GapScore(n□)**: priority-weighted gap closure improvement

This guides retrieval toward actions that are both **informationally optimal** and **computationally efficient**, in contrast to flat or purely recall-oriented selection.

4 MCTSRAG-KG FRAMEWORK

We present MCTSRAG-KG, a modular framework that extends traditional Retrieval-Augmented Generation (RAG) through three key innovations:
(1) gap-aware Monte Carlo Tree Search (MCTS) to prioritize critical retrievals,
(2) orchestrated multi-agent reasoning for refinement and coordination, and
(3) persistent knowledge graphs for transparent fact tracking and provenance. Figure 2 illustrates the end-to-end system.

Each query is decomposed and processed in a structured sequence of steps involving four specialized agents (Planner, Retriever, Validator, Synthesizer) operating on a persistent evidence memory **G=(V,E,L)**. The system iteratively explores the space of candidate retrieval paths using MCTS, with each action (expansion or move) guided by an estimate of its potential to minimize epistemic gaps-e.g., missing entities, relations, or facts that contradict each other-and maximize answer quality.

4.1 Gap-Aware MCTS for Guided Retrieval

Traditional RAG systems rely on static or relevance-based retrievals. MCTSRAG-KG instead formulates retrieval as a sequential decision process guided by a modified UCT formula:

$$\mathrm{UCT}_{\mathrm{gap}}(j) = Q(n_j)/N(n_j) + c\sqrt{\ln N(\mathrm{parent})/N(n_j)} + \alpha \cdot \mathrm{GapScore}(n_j)$$

Here, Q(nj) is the cumulative reward, N(nj) the visit count, c the exploration constant, and GapScore(nj) prioritizes retrievals that close high-impact information gaps. For each node, candidate actions (refine, query, expand, resolve) are evaluated based on both novelty and their ability to reduce prioritized gaps $G_{gap} \subseteq G$.

Reward propagation follows a hybrid utility function:

$$R(n) = \beta \cdot \Delta Gap + (1-\beta) \cdot Quality$$

with Quality derived from coherence, consistency, and source credibility scores [2][11][18].

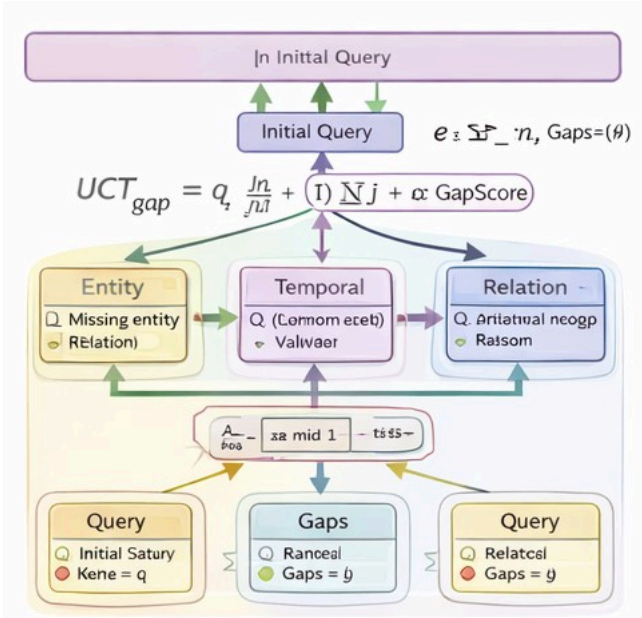

Fig. 2: Example of MCTS-search tree for gap-aware retrieval. Nodes represent retrieval states, with values indicating different types of information gaps. The UCT with gap_score guides exploration, and rewards for resolving gaps are backpropagated up.

4.2 Multi-Agent Reasoning and Coordination

Inspired by MAIN-RAG [8] and MA-RAG [7], the following specialization of agent is used:

- **Planner** segments the query by means of chain-of-thought planning to identify critical subgoals and potential conflicts[16].
- **Retriever** uses MCTS to retrieve evidence without redundancy and traces history using multiple rollouts[1][24].
- **Validator** uses tri-level credibility scoring: reputation (source trust), coherence (semantic consistency), and contradiction detection [11][12][18].
- **Synthesizer** takes the role to aggregate verified content through evidence-aware reasoning mechanisms.

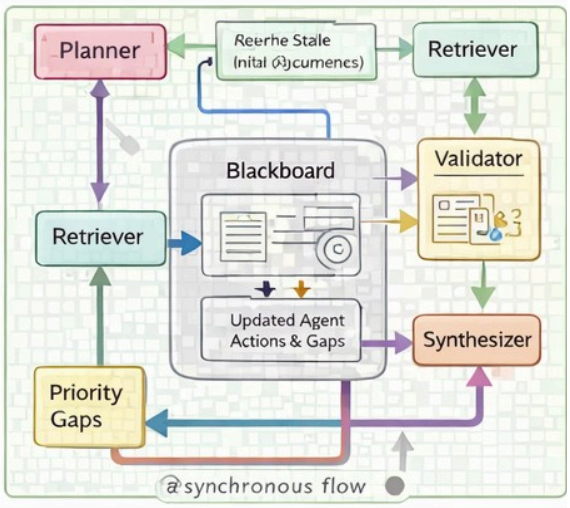

Fig. 3. Multi-agent orchestration loop. Specialized a gents--Planner, Retriever, Validator, and Synthesizer-collaborate by passing messages on detected gaps, re trieved evidence, credibility scores, and updated knowl edge graphs.

Agents coordinate via a message-passing protocol (M) with priority queues, enabling asynchronous blackboard-style communication. Each agent's refinement depends on dynamically updated gap priorities and the evolving knowledge graph (G).

4.3 Persistent Knowledge Graph Memory

We maintain (G = (V, E, L)) as a structured, persistent graph where each new fact node (v \in V) is annotated with {subject, predicate, object, confidence, timestamp, source}. Edges (E) encode semantic relations, support links, or contradictions. Gaps (G_{\text{gap}} \subseteq V) are labeled by type (e.g., missing entity, relation, conflict) and assigned priorities (p \in [0,1]) and resolution states.

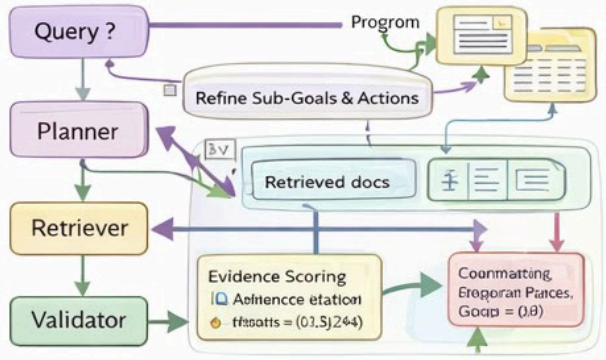

Fig. 4. Persistent knowledge graph construction and evidence chaining. As facts are validated, they are incorporated into the knowledge graph. Evidence chains link facts with labels like "supports", "refutes", and qualify facts with confidence, sources.

This KG design supports provenance-aware reasoning [13][26], conflict monitoring [11], and dynamic context expansion across retrieval iterations [27].

4.4 Integrated Retrieval Algorithm

Algorithm 1: MCTS-Guided RAG with KG

Input: Query q, source set S, max iterations T, exploration constant c
Output: Answer A, Evidence Chains E, gap report R

```
Initialize KG: G ← (V, E, L)
root ← (∅, q, ∅, G
gaps ← Planner.identify_gaps(q)
for t = 1 to T do
  // Tree Search
  current ← root
  while current not in terminals:
      current ← max_child(current, UCT_gap)
  // Candidate Evaluation
  actions ← generate_actions(current,gaps)
  for a in actions do
      new ← execute(a, current)
      docs ← retrieve(a.query, S)
      validated ← Validator.assess(docs)
      facts ← extract(validated)
      G ← update_KG(G, facts)
      ΔGap ← measure_gap_reduction(G)
  q_score ← evaluate_quality(validated)
```

```
    reward ← β * δGap + (1–β) * q_score
    backpropagate(new, reward)
if termination_condition_met: break
A ← Synthesizer.generate(G)
E ← extract_evidence_chains(G)
R ← report_gaps(G)
return A, E, R
```

4.5 Gap-Driven Reward Function

To direct the retrieval towards the most informative documents, we specify a weighted reward function:

$$\text{GapScore}(n) = \sum_{g \in G_{gap}} w(g_{type}) \cdot \text{priority}(g) \cdot \text{resolved}(g,n)$$

Weights $w(g_{type})$ reflect gap type importance (e.g., entities > conflicts > temporal) [29], and resolved $(g, n) \in \{0,1\}$ indicates if the gap is addressed at node n. This mechanism encourages retrievals that minimize uncertainty and hallucination [12][20].

5 DISCUSSION AND CONCLUSION

MCTSRAG-KG addresses core limitations in RAG by integrating structured decision-making, coordinated reasoning, and transparent memory. Our contributions include:

- **Gap-Aware Retrieval**: Extends MCTS-RAG [2] with structured gap tracking and explicit reward prioritization, enabling systematic closure of information gaps over redundant retrievals.
- **Multi-Agent Specialization**: Inspired by MA-RAG [7], MAIN-RAG [8], and MAgICoRe [9], our architecture decomposes retrieval tasks into coherent subtasks, improving factuality and scalability.
- **Knowledge Graph Memory**: Supports long-context reasoning [13][26], evidence provenance [11], and persistent fact tracking beyond session limits.

Practical Implications

The approach generalizes to domains requiring:

- Fact-checking and claim verification (e.g., financial reports, journalism).
- Temporal reasoning and simulation (e.g., crisis forecasting).
- Legal/regulatory compliance (traceable evidence chains).
- Multi-round research and retrieval assistant pipelines.

Limitations

- **MCTS Overhead**: Computationally expensive (addressed via learned heuristics [22][29]).
- **KG Scalability**: Extracting and maintaining large KGs across long queries remains a challenge [27].
- **Heuristic Dependency**: Gap detection depends on accurate NER/linking [13][14].

Future Work

- Integration of Tree-of-Thoughts prompting within agents [17].
- Reinforcement learning from user feedback to optimize reward shaping [23].
- Richer multi-modal reasoning using structured graphs, tables, and charts [25].

- Domain-adaptive collaborative KG building across users.

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